Kirolos Sedra

RAN & Cellular Systems Engineer

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Professional Summary

RAN and cellular systems engineer with hands-on experience integrating and validating private **5G SA** networks across the gNB, 5G core, SDR, and UE. Experienced in UE registration and PDU-session debugging, RF and mobility testing, cellular KPI analysis, and end-to-end diagnostics using **srsRAN**, **Open5GS**, **USRP**, and Quectel modems.

Experience

Wireless Sensors and Devices Lab, University of Waterloo **Sep 2024 – Present**
Research Assistant *Full-time, On-site*

- Built and validated a private **5G SA testbed** using an **srsRAN gNB**, **Open5GS 5GC**, **USRP B210**, and **Quectel RM502Q-AE UE**; resolved AMF/SMF/UPF registration, PDU-session, and user-plane connectivity issues.
- Executed attenuation-based SNR ladders, static OTA tests, and mobility scenarios; measured **UL/DL throughput**, latency under load, **MCS**, **CQI**, **RSRP**, **RSRQ**, and **SINR**, and investigated asymmetry through gNB, core, AT, and QMI logs.
- Characterized indoor and outdoor **5G/LTE and Wi-Fi performance for mobile-robot teleoperation**, correlating radio-quality measurements with P99 latency and throughput to quantify mobility, interference, and LTE-anchoring effects.
- Evaluated **NR-U/Wi-Fi coexistence** using ns-3.35, CTTC 5G-LENA, and NR-U extensions, analyzing throughput, latency, jitter, fairness, Wi-Fi starvation, and NR penalties under varied LBT behavior and traffic load.
- Automated wireless validation for up to **30 DUTs**, reducing manual intervention by approximately **70%** through RF logging, network-service automation, SSH-based configuration, and PHY/transport analysis.

Aquasensing **Sep 2024 – Present**
Firmware Specialist / Research Intern *Part-time*

- Validated and debugged production **BLE–LTE IoT gateways** through RF-informed and anechoic-chamber testing, component stubs, and cross-layer isolation of firmware, connectivity, and cloud failures.
- Managed cellular IoT connectivity through **Hologram MVNO** and **Bell Jasper**; debugged **nRF/Zephyr** firmware and automated provisioning and validation workflows for **70 IoT gateways**.

University of Waterloo **Feb 2025 – Apr 2026**
Teaching Assistant, ECE 610 Broadband Networks and ECE 358 Computer Networks *Part-time*

Ejada Systems **Sep 2023 – Aug 2024**
Software Engineer – Java Spring Boot, REST APIs, SQL Optimization *Full-time*

Publications

CCECE 2026: Trajectory-Based Performance Characterization of Wi-Fi for Outdoor Mobile Robot Teleoperation; Scalability Analysis of Wi-Fi Broadcast Mechanisms Using Simulation and Hybrid Approaches.

Education

University of Waterloo – Master of Applied Science, Electrical and Computer Engineering Sep 2024 – Aug 2026
Grade: 92% (A+)

Ain Shams University – Bachelor of Science, Computer Engineering Sep 2018 – Jul 2023
CGPA: 3.12/4.00

Technical Skills

Cellular/RAN: 5G NR SA/NSA, LTE, RAN architecture, PHY/MAC/RRC procedures, UE registration, PDU sessions, UL/DL analysis, mobility testing, 3GPP TS 38-series exposure
Platforms/Diagnostics: srsRAN, Open5GS, USRP B210, Quectel modems, AT commands, QMI, gNB/5GC logs, Wireshark
RF/Performance: RSRP, RSRQ, SINR, SNR, CQI, MCS, throughput, latency under load, attenuators, OTA, spectrum analyzer, VNA, anechoic chambers, interference/coexistence
Simulation/Systems: ns-3, CTTC 5G-LENA, NR-U, GNU Radio, MATLAB, Simulink, Python, C/C++, Linux, TCP/IP, SSH